

Printer Dover

Spruce version 11.2 -- spooler version 11.2

File: MemX-Rev-Cg.sil etc.

Creation date: 29-Oct-79 14:31:55

For: Spruce

26 total sheets = 25 pages, 1 copy.

Problems encountered:

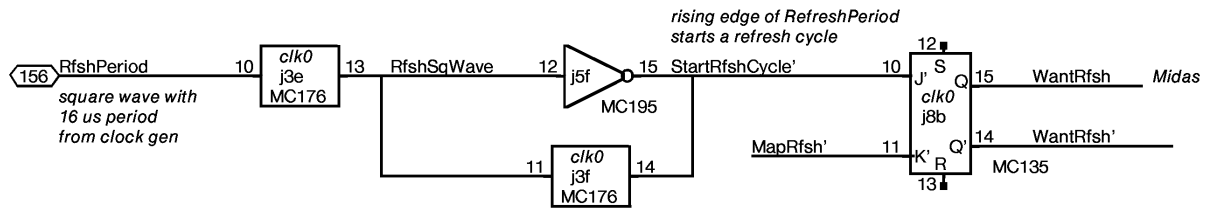
Font HELVETICA24B substituted for font HELVE24.

DORADO SCHEMATICS

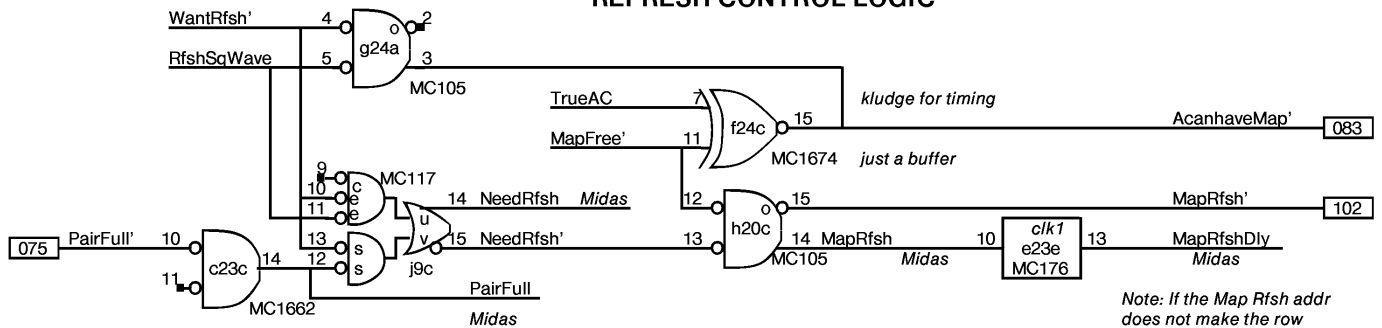
Mem X

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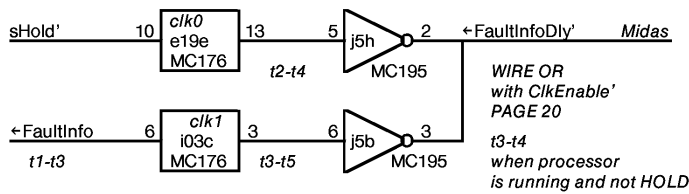
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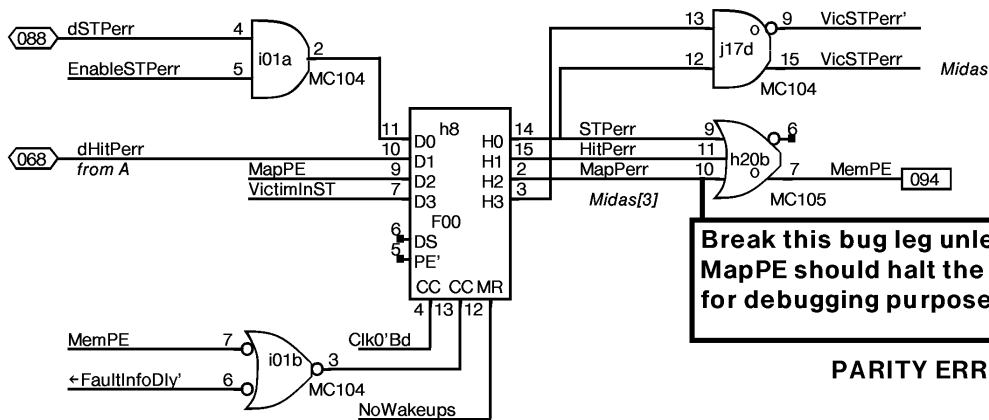
REFRESH CONTROL LOGIC



← FAULTINFO LOGIC



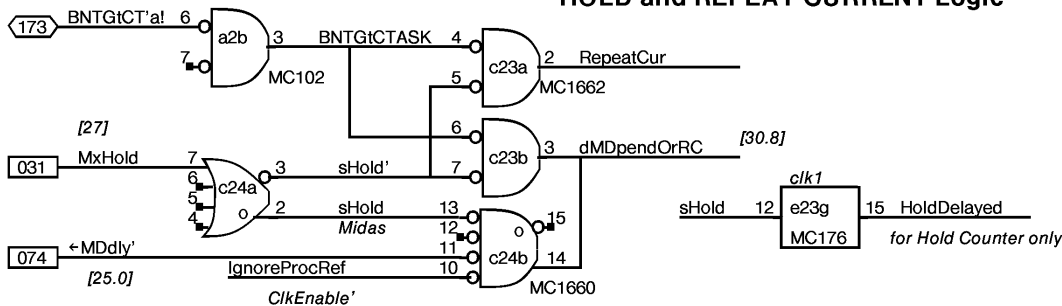
Note: If the Map Rfsh addr does not make the row address hold time, add a clk0 FF in series with this clk1 FF and OR the outputs with an OR gate.



Break this bug leg unless MapPE should halt the Dorado for debugging purposes

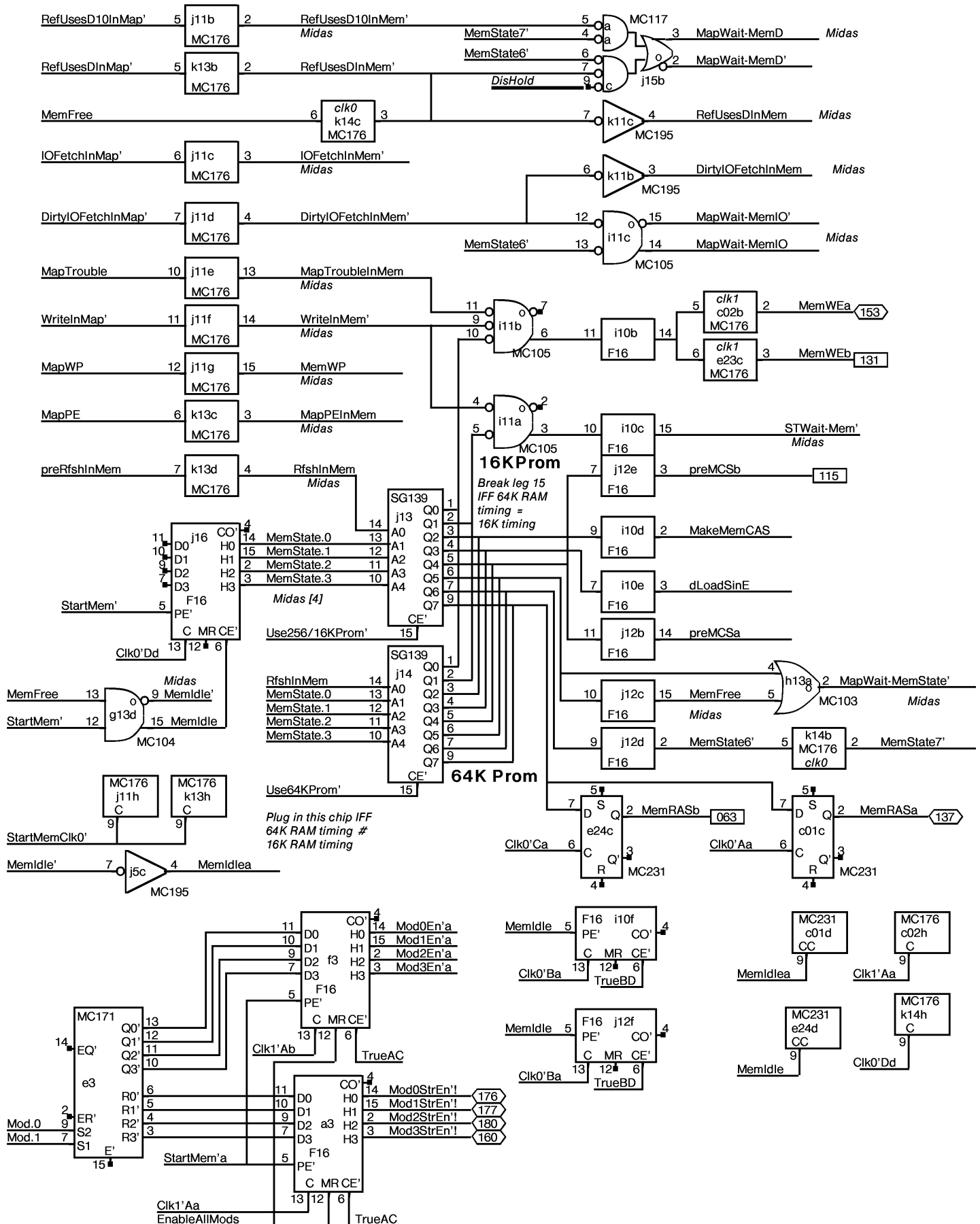
PARITY ERROR LOGIC

HOLD and REPEAT CURRENT Logic

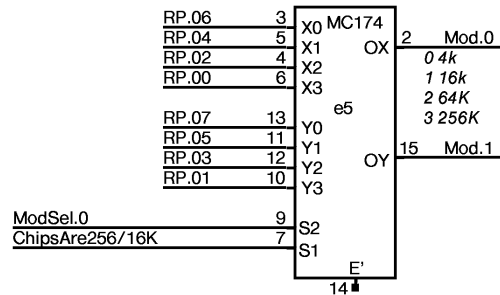




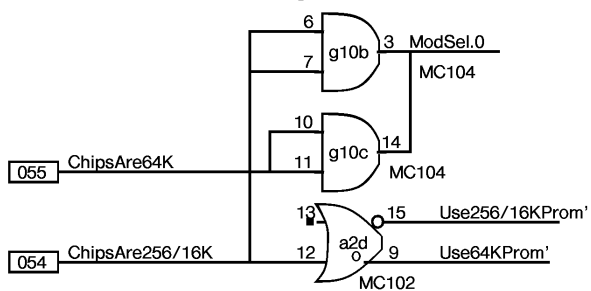




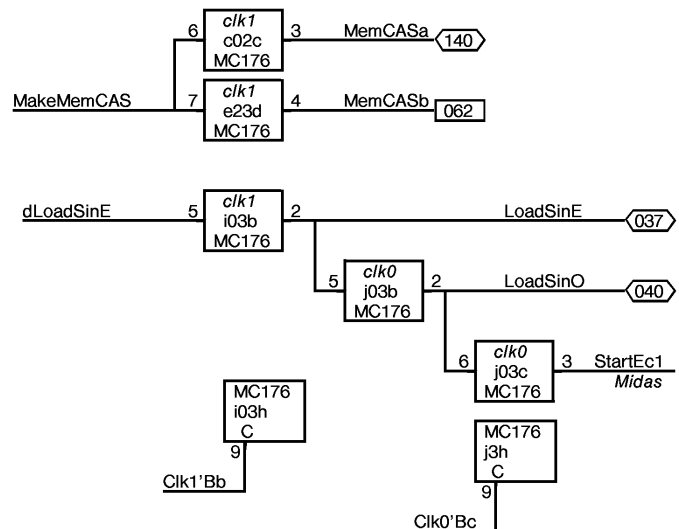
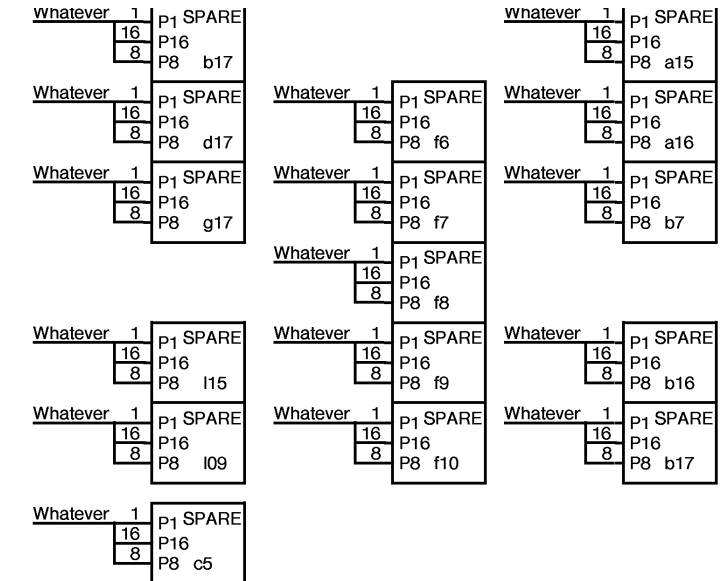
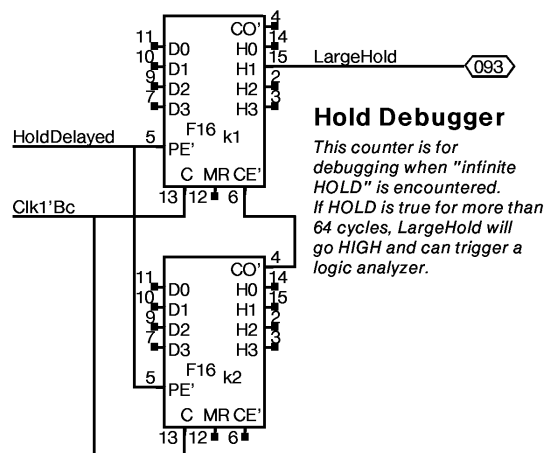
Only one of the ChipsAreXXX signals should be true at any time.

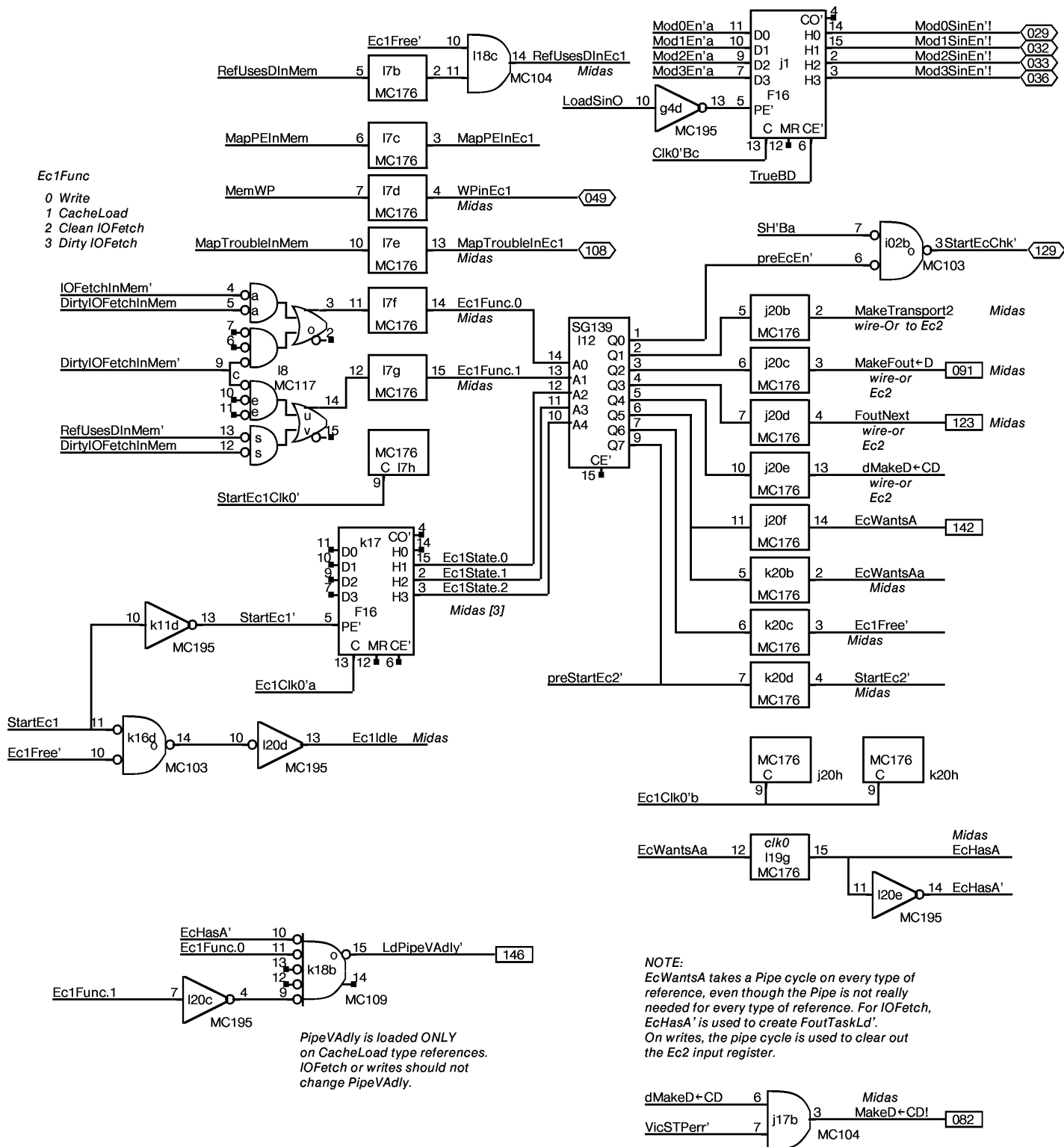


**BREAK THIS BUG LEG
UNTIL 256K CHIPS
ARE USED**

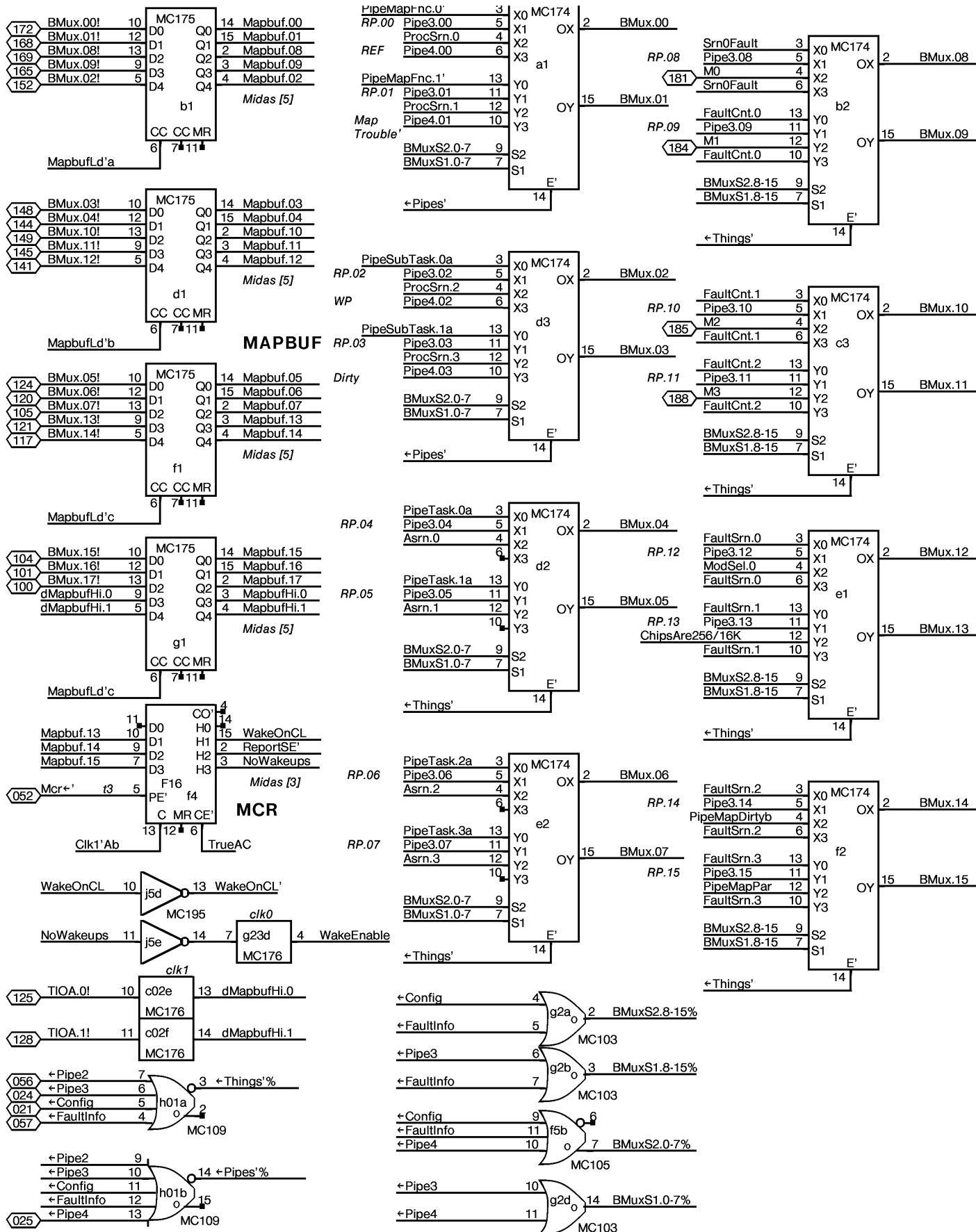


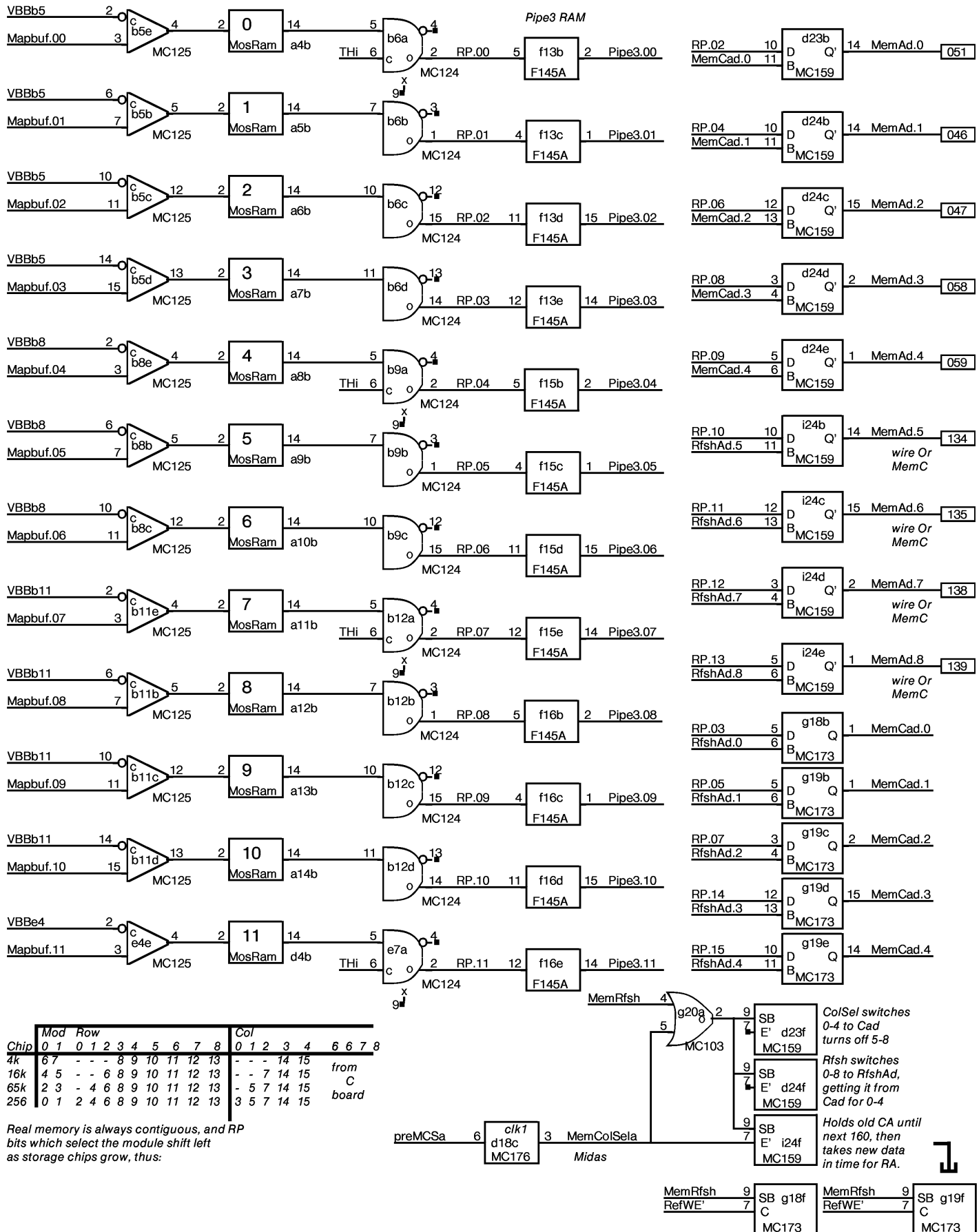
When the only two chip options for Dorado are 64K or 256K, then interpret this signal as ChipsAre256K and enable the upper gate of ModSel.0

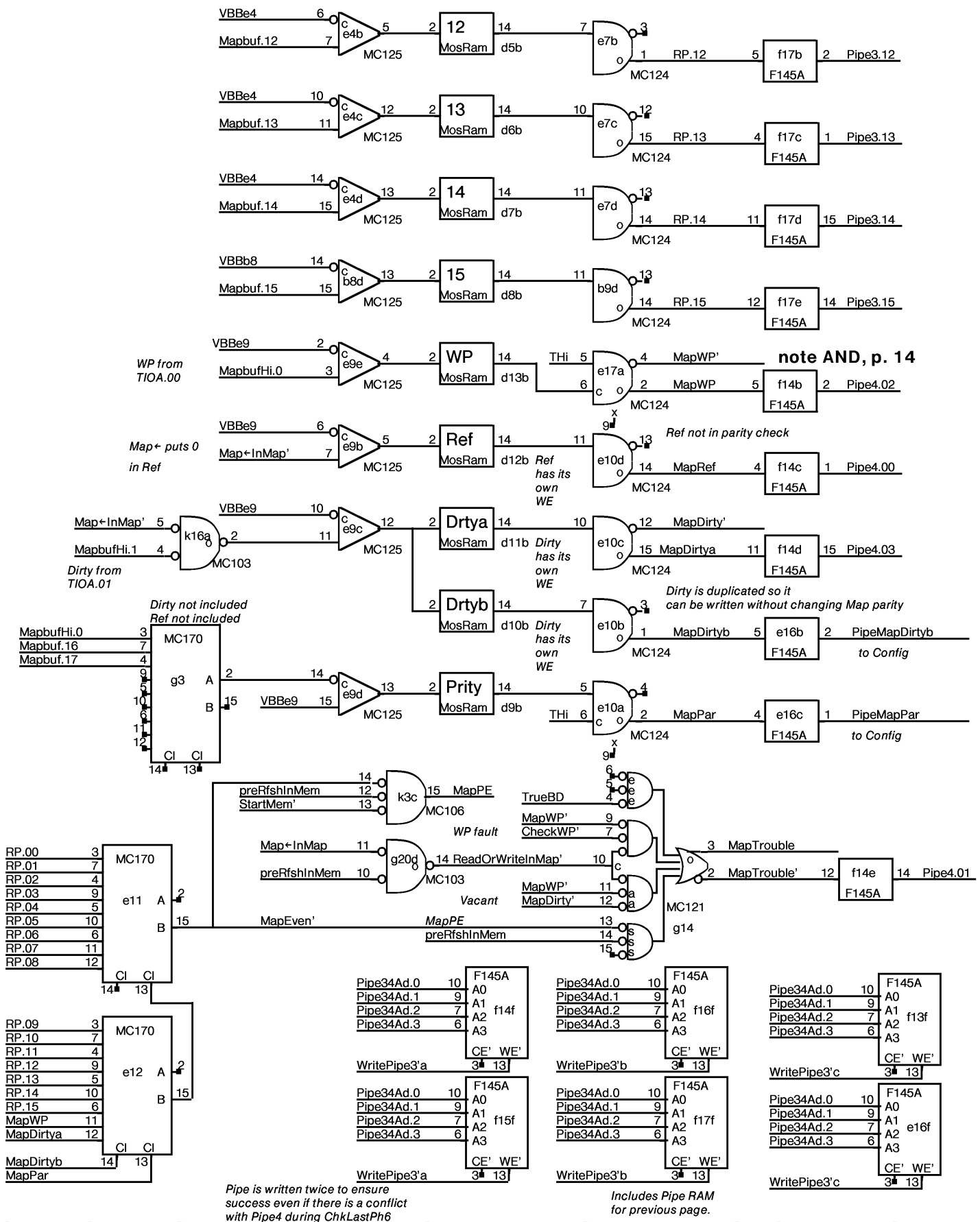


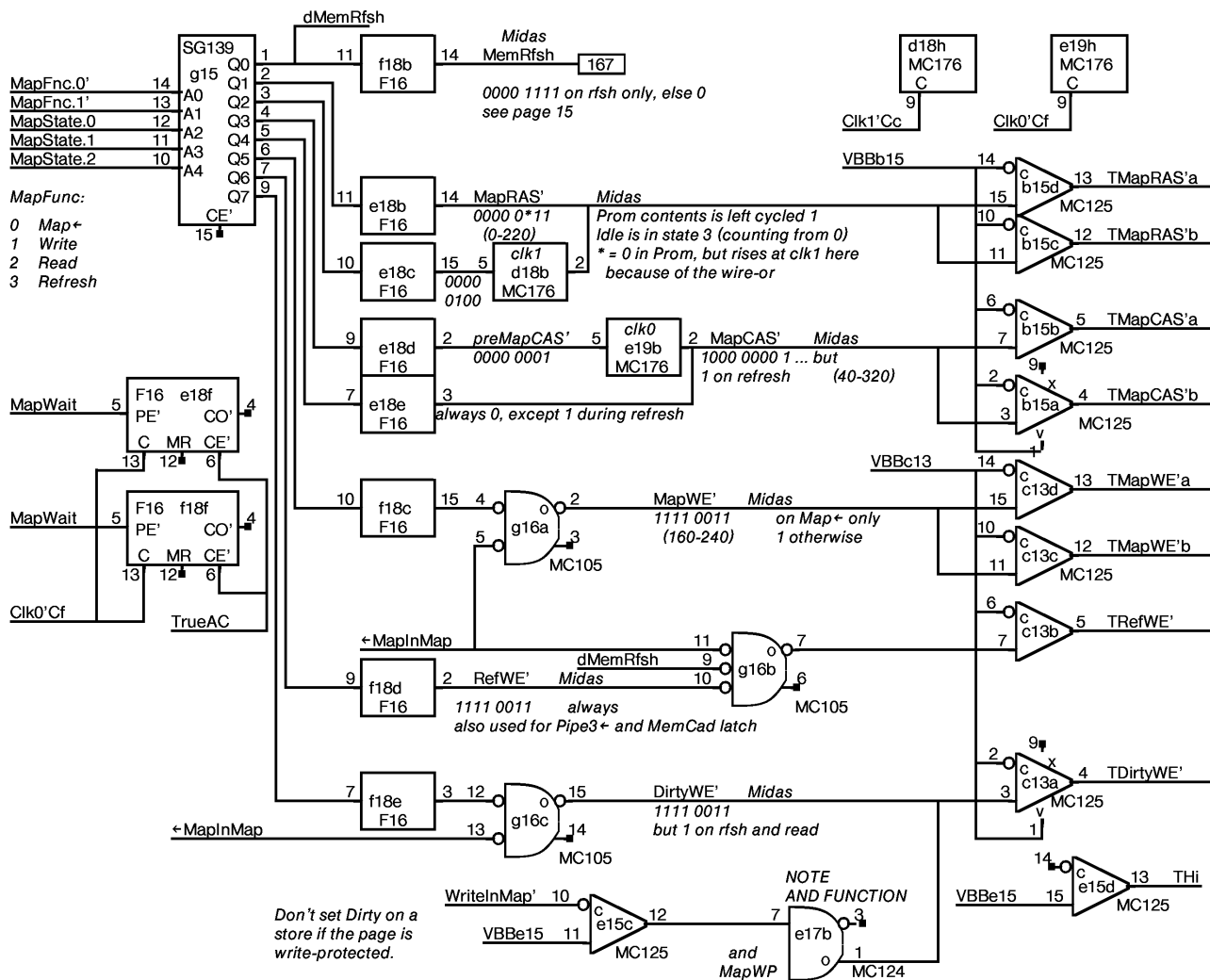










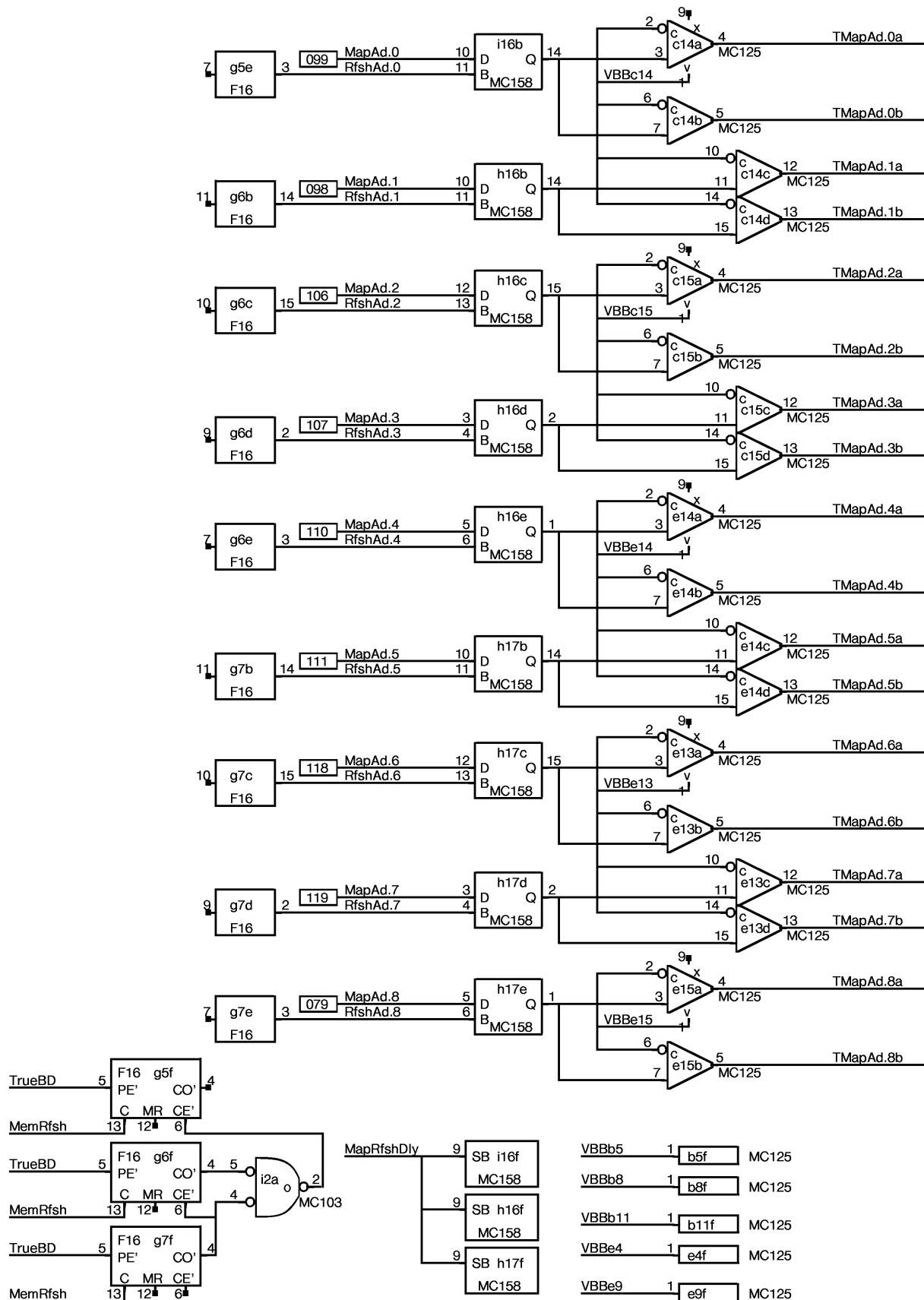


PLATS are BECKMAN 898-3-R027 series resistors 8 per chip

TMapAd.0a	1	P1	d16	P16	16	RTMapAd.0a
TMapAd.1a	2	P2	P15		15	RTMapAd.1a
TMapAd.2a	3	P3	P14		14	RTMapAd.2a
TMapAd.3a	4	P4	P13		13	RTMapAd.3a
TMapAd.4a	5	P5	P12		12	RTMapAd.4a
TMapAd.5a	6	P6	P11		11	RTMapAd.5a
TMapAd.6a	7	P7	P10		10	RTMapAd.6a
TMapAd.7a	8	P8	P9		9	RTMapAd.7a
PLAT1816						
TMapAd.8a	1	P1	d15	P16	16	RTMapAd.8a
TMapWE'a	2	P2	P15		15	RTMapWE'a
TMapRAS'a	3	P3	P14		14	RTMapRAS'a
TMapCAS'a	4	P4	P13		13	RTMapCAS'a
TMapCAS'b	5	P5	P12		12	RTMapCAS'b
TMapRAS'b	6	P6	P11		11	RTMapRAS'b
TMapWE'b	7	P7	P10		10	RTMapWE'b
TMapAd.0b	8	P8	P9		9	RTMapAd.0b
PLAT1816						
TMapAd.1b	1	P1	d14	P16	16	RTMapAd.1b
TMapAd.2b	2	P2	P15		15	RTMapAd.2b
TMapAd.3b	3	P3	P14		14	RTMapAd.3b
TMapAd.4b	4	P4	P13		13	RTMapAd.4b
TMapAd.5b	5	P5	P12		12	RTMapAd.5b
TMapAd.6b	6	P6	P11		11	RTMapAd.6b
TMapAd.7b	7	P7	P10		10	RTMapAd.7b
TMapAd.8b	8	P8	P9		9	RTMapAd.8b
PLAT1816						

PLATS are Jumper wires for configuration

VCC	1	P1	P16	16	Whatever
RTMapAd.0a	2	P2	P15	15	RamA0orVEEaI01
VEE	3	P3	P14	14	
RTMapAd.1a	4	P4	P13	13	RamA1orVCCaI01
VCC	5	P5	P12	12	
VCC	6	P6	P11	11	RamV + aI01
VDD	7	P7	P10	10	
Whatever	8	P8	P9	9	Whatever
PLAT					
VCC	1	P1	P16	16	Whatever
RTMapAd.0b	2	P2	P15	15	RamA0orVEEbI01
VEE	3	P3	P14	14	
RTMapAd.1b	4	P4	P13	13	RamA1orVCCbI01
VCC	5	P5	P12	12	
VCC	6	P6	P11	11	RamV + bI01
VDD	7	P7	P10	10	
Whatever	8	P8	P9	9	Whatever
PLAT					
Two identical platforms.					
Whatever	1	P1	P16	16	Whatever
RamV + aI02	2	P2	P15	15	GND
	3	P3	P14	14	
	4	P4	P13	13	
RamA1orVCCaI02	5	P5	P12	12	
RamA0orVEEaI02	6	P6	P11	11	
	7	P7	P10	10	
Whatever	8	P8	P9	9	Whatever
PLAT					
Whatever	1	P1	P16	16	Whatever
RamV + bI02	2	P2	P15	15	GND
	3	P3	P14	14	
	4	P4	P13	13	
RamA1orVCCbI02	5	P5	P12	12	
RamA0orVEEbI02	6	P6	P11	11	
	7	P7	P10	10	
Whatever	8	P8	P9	9	Whatever
PLAT					



0000 1111 on rfsh only, else 0

RamA0orVEEa115	1	A0
RamA1orVCCa114	9	A1
RTMapAd.2a	5	A2 MosRam
RTMapAd.3a	7	A3
RTMapAd.4a	6	A4
RTMapAd.5a	12	A5
RTMapAd.6a	11	A6
RTMapAd.7a	10	A7
RTMapAd.8a	13	A8
RTMapRAS'a	4	RAS'
RTMapCAS'a	15	CAS'
RTMapWE'a	3	WE'
RamV + a118	8	V +
GND	16	GRND

RP.00

a4

RamA0orVEEa109	1	A0
RamA1orVCCa109	9	A1
RTMapAd.2a	5	A2 MosRam
RTMapAd.3a	7	A3
RTMapAd.4a	6	A4
RTMapAd.5a	12	A5
RTMapAd.6a	11	A6
RTMapAd.7a	10	A7
RTMapAd.8a	13	A8
RTMapRAS'a	4	RAS'
RTMapCAS'a	15	CAS'
RTMapWE'a	3	WE'
RamV + a111	8	V +
GND	16	GRND

RP.05

a9

RamA0orVEEa103	1	A0
RamA1orVCCa103	9	A1
RTMapAd.2a	5	A2 MosRam
RTMapAd.3a	7	A3
RTMapAd.4a	6	A4
RTMapAd.5a	12	A5
RTMapAd.6a	11	A6
RTMapAd.7a	10	A7
RTMapAd.8a	13	A8
RTMapRAS'a	4	RAS'
RTMapCAS'a	15	CAS'
RTMapWE'a	3	WE'
RamV + a103	8	V +
GND	16	GRND

RP.10

a14

RamA0orVEEa114	1	A0
RamA1orVCCa113	9	A1
RTMapAd.2a	5	A2 MosRam
RTMapAd.3a	7	A3
RTMapAd.4a	6	A4
RTMapAd.5a	12	A5
RTMapAd.6a	11	A6
RTMapAd.7a	10	A7
RTMapAd.8a	13	A8
RTMapRAS'a	4	RAS'
RTMapCAS'a	15	CAS'
RTMapWE'a	3	WE'
RamV + a117	8	V +
GND	16	GRND

RP.01

a5

RamA0orVEEa108	1	A0
RamA1orVCCa107	9	A1
RTMapAd.2a	5	A2 MosRam
RTMapAd.3a	7	A3
RTMapAd.4a	6	A4
RTMapAd.5a	12	A5
RTMapAd.6a	11	A6
RTMapAd.7a	10	A7
RTMapAd.8a	13	A8
RTMapRAS'a	4	RAS'
RTMapCAS'a	15	CAS'
RTMapWE'a	3	WE'
RamV + a109	8	V +
GND	16	GRND

RP.06

a10

GND	5	SIP	b42f
RamV + a119	8	SIP	b42i
GND	4	SIP	b42e
RamA0orVEEa116	1	SIP	b42b

GND	5	SIP	b43f
RamV + a116	8	SIP	b43i
GND	4	SIP	b43e
RamA1orVCCa115	1	SIP	b43b

RamA0orVEEa113	1	A0
RamA1orVCCa112	9	A1
RTMapAd.2a	5	A2 MosRam
RTMapAd.3a	7	A3
RTMapAd.4a	6	A4
RTMapAd.5a	12	A5
RTMapAd.6a	11	A6
RTMapAd.7a	10	A7
RTMapAd.8a	13	A8
RTMapRAS'a	4	RAS'
RTMapCAS'a	15	CAS'
RTMapWE'a	3	WE'
RamV + a115	8	V +
GND	16	GRND

RP.02

a6

RamA0orVEEa107	1	A0
RamA1orVCCa106	9	A1
RTMapAd.2a	5	A2 MosRam
RTMapAd.3a	7	A3
RTMapAd.4a	6	A4
RTMapAd.5a	12	A5
RTMapAd.6a	11	A6
RTMapAd.7a	10	A7
RTMapAd.8a	13	A8
RTMapRAS'a	4	RAS'
RTMapCAS'a	15	CAS'
RTMapWE'a	3	WE'
RamV + a108	8	V +
GND	16	GRND

RP.07

a11

GND	5	SIP	b44f
RamV + a113	8	SIP	b44i
GND	4	SIP	b44e
RamA0orVEEa111	1	SIP	b44b

GND	5	SIP	b45f
RamV + a110	8	SIP	b45i
GND	4	SIP	b45e
RamA1orVCCa108	1	SIP	b45b

RamA0orVEEa112	1	A0
RamA1orVCCa111	9	A1
RTMapAd.2a	5	A2 MosRam
RTMapAd.3a	7	A3
RTMapAd.4a	6	A4
RTMapAd.5a	12	A5
RTMapAd.6a	11	A6
RTMapAd.7a	10	A7
RTMapAd.8a	13	A8
RTMapRAS'a	4	RAS'
RTMapCAS'a	15	CAS'
RTMapWE'a	3	WE'
RamV + a114	8	V +
GND	16	GRND

RP.03

a7

RamA0orVEEa105	1	A0
RamA1orVCCa106	9	A1
RTMapAd.2a	5	A2 MosRam
RTMapAd.3a	7	A3
RTMapAd.4a	6	A4
RTMapAd.5a	12	A5
RTMapAd.6a	11	A6
RTMapAd.7a	10	A7
RTMapAd.8a	13	A8
RTMapRAS'a	4	RAS'
RTMapCAS'a	15	CAS'
RTMapWE'a	3	WE'
RamV + a106	8	V +
GND	16	GRND

RP.08

a12

GND	5	SIP	b46f
RamV + a107	8	SIP	b46i
GND	4	SIP	b46e
RamA0orVEEa106	1	SIP	b46b

GND	5	SIP	b47f
RamV + a104	8	SIP	b47i
GND	4	SIP	b47e
RamA1orVCCa104	1	SIP	b47b

RamA0orVEEa110	1	A0
RamA1orVCCa110	9	A1
RTMapAd.2a	5	A2 MosRam
RTMapAd.3a	7	A3
RTMapAd.4a	6	A4
RTMapAd.5a	12	A5
RTMapAd.6a	11	A6
RTMapAd.7a	10	A7
RTMapAd.8a	13	A8
RTMapRAS'a	4	RAS'
RTMapCAS'a	15	CAS'
RTMapWE'a	3	WE'
RamV + a112	8	V +
GND	16	GRND

RP.04

a8

RamA0orVEEa104	1	A0
RamA1orVCCa105	9	A1
RTMapAd.2a	5	A2 MosRam
RTMapAd.3a	7	A3
RTMapAd.4a	6	A4
RTMapAd.5a	12	A5
RTMapAd.6a	11	A6
RTMapAd.7a	10	A7
RTMapAd.8a	13	A8
RTMapRAS'a	4	RAS'
RTMapCAS'a	15	CAS'
RTMapWE'a	3	WE'
RamV + a105	8	V +
GND	16	GRND

RP.09

a13

Capacitor Options:

16K RAMs: install CAPS
between pins 5 and 8
AND between pins 1 and 4,
b42,b43,b44,b45,b46,b47

64K RAMs: install CAPS
between pins 5 and 8 ONLY,
b42,b43,b44,b45,b46,b47

256K RAMs: same as 64K.

RamA0orVEEb!14	1	A0
RamA1orVCCb!14	9	A1
RTMapAd.2b	5	A2 MosRam
RTMapAd.3b	7	A3
RTMapAd.4b	6	A4
RTMapAd.5b	12	A5
RTMapAd.6b	11	A6
RTMapAd.7b	10	A7
RTMapAd.8b	13	A8
RTMapRAS'b	4	RAS'
RTMapCAS'b	15	CAS'
RTMapWE'b	3	WE'
RamV + b!17	8	V +
GND	16	GRND

RP.11

d4

RamA0orVEEb!08	1	A0
RamA1orVCCb!09	9	A1
RTMapAd.2b	5	A2 MosRam
RTMapAd.3b	7	A3
RTMapAd.4b	6	A4
RTMapAd.5b	12	A5
RTMapAd.6b	11	A6
RTMapAd.7b	10	A7
RTMapAd.8b	13	A8
RTMapRAS'b	4	RAS'
RTMapCAS'b	15	CAS'
RTMapWE'b	3	WE'
RamV + b!10	8	V +
GND	16	GRND

Parity

d9

RamA0orVEEb!13	1	A0
RamA1orVCCb!13	9	A1
RTMapAd.2b	5	A2 MosRam
RTMapAd.3b	7	A3
RTMapAd.4b	6	A4
RTMapAd.5b	12	A5
RTMapAd.6b	11	A6
RTMapAd.7b	10	A7
RTMapAd.8b	13	A8
RTMapRAS'b	4	RAS'
RTMapCAS'b	15	CAS'
RTMapWE'b	3	WE'
RamV + b!16	8	V +
GND	16	GRND

RP.12

d5

RamA0orVEEb!07	1	A0
RamA1orVCCb!07	9	A1
RTMapAd.2b	5	A2 MosRam
RTMapAd.3b	7	A3
RTMapAd.4b	6	A4
RTMapAd.5b	12	A5
RTMapAd.6b	11	A6
RTMapAd.7b	10	A7
RTMapAd.8b	13	A8
RTMapRAS'b	4	RAS'
RTMapCAS'b	15	CAS'
TDirtyWE'	3	WE'
RamV + b!08	8	V +
GND	16	GRND

DirtyB

d10

GND	5	SIP	d42f
RamV + b!18	8	SIP	d42i
GND	4	SIP	d42e
RamA0orVEEb!15	1	SIP	d42b

GND	5	SIP	d43f
RamV + b!15	8	SIP	d43i
GND	4	SIP	d43e
RamA1orVCCb!15	1	SIP	d43b

RamA0orVEEb!12	1	A0
RamA1orVCCb!12	9	A1
RTMapAd.2b	5	A2 MosRam
RTMapAd.3b	7	A3
RTMapAd.4b	6	A4
RTMapAd.5b	12	A5
RTMapAd.6b	11	A6
RTMapAd.7b	10	A7
RTMapAd.8b	13	A8
RTMapRAS'b	4	RAS'
RTMapCAS'b	15	CAS'
RTMapWE'b	3	WE'
RamV + b!14	8	V +
GND	16	GRND

RP.13

d6

RamA0orVEEb!06	1	A0
RamA1orVCCb!06	9	A1
RTMapAd.2b	5	A2 MosRam
RTMapAd.3b	7	A3
RTMapAd.4b	6	A4
RTMapAd.5b	12	A5
RTMapAd.6b	11	A6
RTMapAd.7b	10	A7
RTMapAd.8b	13	A8
RTMapRAS'b	4	RAS'
RTMapCAS'b	15	CAS'
TDirtyWE'	3	WE'
RamV + b!07	8	V +
GND	16	GRND

DirtyA

d11

GND	5	SIP	d44f
RamV + b!12	8	SIP	d44i
GND	4	SIP	d44e
RamA0orVEEb!10	1	SIP	d44b

GND	5	SIP	d45f
RamV + b!09	8	SIP	d45i
GND	4	SIP	d45e
RamA1orVCCb!08	1	SIP	d45b

RamA0orVEEb!11	1	A0
RamA1orVCCb!11	9	A1
RTMapAd.2b	5	A2 MosRam
RTMapAd.3b	7	A3
RTMapAd.4b	6	A4
RTMapAd.5b	12	A5
RTMapAd.6b	11	A6
RTMapAd.7b	10	A7
RTMapAd.8b	13	A8
RTMapRAS'b	4	RAS'
RTMapCAS'b	15	CAS'
RTMapWE'b	3	WE'
RamV + b!13	8	V +
GND	16	GRND

RP.14

d7

RamA0orVEEb!04	1	A0
RamA1orVCCb!05	9	A1
RTMapAd.2b	5	A2 MosRam
RTMapAd.3b	7	A3
RTMapAd.4b	6	A4
RTMapAd.5b	12	A5
RTMapAd.6b	11	A6
RTMapAd.7b	10	A7
RTMapAd.8b	13	A8
RTMapRAS'b	4	RAS'
RTMapCAS'b	15	CAS'
TRefWE'	3	WE'
RamV + b!05	8	V +
GND	16	GRND

Ref

d12

GND	5	SIP	d46f
RamV + b!06	8	SIP	d46i
GND	4	SIP	d46e
RamA0orVEEb!05	1	SIP	d46b

GND	5	SIP	d47f
RamV + b!04	8	SIP	d47i
GND	4	SIP	d47e
RamA1orVCCb!04	1	SIP	d47b

RamA0orVEEb!09	1	A0
RamA1orVCCb!10	9	A1
RTMapAd.2b	5	A2 MosRam
RTMapAd.3b	7	A3
RTMapAd.4b	6	A4
RTMapAd.5b	12	A5
RTMapAd.6b	11	A6
RTMapAd.7b	10	A7
RTMapAd.8b	13	A8
RTMapRAS'b	4	RAS'
RTMapCAS'b	15	CAS'
RTMapWE'b	3	WE'
RamV + b!11	8	V +
GND	16	GRND

RP.15

d8

RamA0orVEEb!03	1	A0
RamA1orVCCb!03	9	A1
RTMapAd.2b	5	A2 MosRam
RTMapAd.3b	7	A3
RTMapAd.4b	6	A4
RTMapAd.5b	12	A5
RTMapAd.6b	11	A6
RTMapAd.7b	10	A7
RTMapAd.8b	13	A8
RTMapRAS'b	4	RAS'
RTMapCAS'b	15	CAS'
RTMapWE'b	3	WE'
RamV + b!03	8	V +
GND	16	GRND

WP

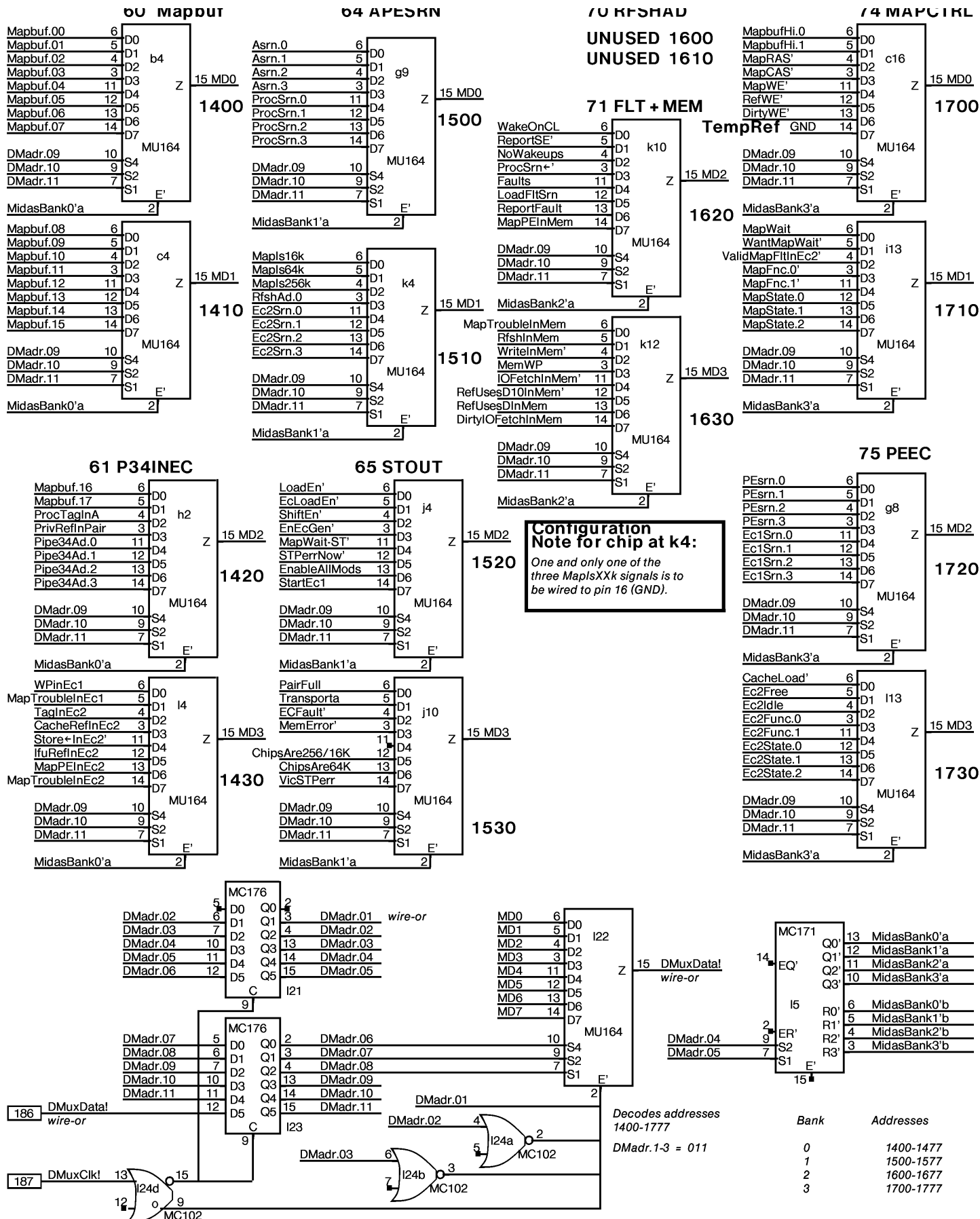
d13

Capacitor Options:

16K RAMs: install CAPS
between pins 5 and 8
AND between pins 1 and 4,
d42,d43,d44,d45,d46,d47

64K RAMs: install CAPS
between pins 5 and 8 ONLY,
d42,d43,d44,d45,d46,d47

256K RAMs: same as 64K.



62 MCDTSK

MDMtagAd.0	6	D0
MDMtagAd.1	5	D1 a19
MDMtagAd.2	4	D2
MDMtagAd.3	3	D3
CurTask.0	11	D4
CurTask.1	12	D5
CurTask.2	13	D6
CurTask.3	14	D7
DMadr.09	10	S4
DMadr.10	9	S2
DMadr.11	7	S1

15 MD4

1440

MidasBank0'b	2	E'
ProcTag	6	D0
MDMtag	5	D1 b18
At = Curt'	4	D2
Dt = Curt'	3	D3
Dtask.0	11	D4
Dtask.1	12	D5
Dtask.2	13	D6
Dtask.3	14	D7
DMadr.09	10	S4
DMadr.10	9	S2
DMadr.11	7	S1

15 MD5

1450

66 TAG + AT

MemColSela	6	D0
EcHasA	5	D1 f20
Ptag	4	D2
MapWait-Ec2	3	D3
Dtag'	11	D4
sHold	12	D5
MapWait-MemState'	13	D6
MapRfsh	14	D7
DMadr.09	10	S4
DMadr.10	9	S2
DMadr.11	7	S1

15 MD4

1540

MidasBank1'b	2	E'
AcanHaveD	6	D0
CacheRefinPair'	5	D1 b20
EcWordRefToD	4	D2
ChkLastPh6	3	D3
Atask.0	11	D4
Atask.1	12	D5
Atask.2	13	D6
Atask.3	14	D7
DMadr.09	10	S4
DMadr.10	9	S2
DMadr.11	7	S1

15 MD5

1550

72 RFS + SRN

STPerr	6	D0
MapPerr	5	D1 k8
HitPerr	4	D2
WantRfsh	3	D3
NeedRfsh	11	D4
StartMema	12	D5
StkWake	13	D6
+ FaultInfoDly'	14	D7
DMadr.09	10	S4
DMadr.10	9	S2
DMadr.11	7	S1

15 MD4

1640

MidasBank2'b	2	E'
MapSrn.0	6	D0
MapSrn.1	5	D1 i21
MapSrn.2	4	D2
MapSrn.3	3	D3
MemSrn.0	11	D4
MemSrn.1	12	D5
MemSrn.2	13	D6
MemSrn.3	14	D7
DMadr.09	10	S4
DMadr.10	9	S2
DMadr.11	7	S1

15 MD5

1650

76 INMAP

RefUsesDInMap'	6	D0
RefUsesD10InMap'	5	D1 h7
DirtyIOFetchInMap'	4	D2
WriteInMap'	3	D3
IOFetchInMap'	11	D4
+ MapInMap'	12	D5
Store+InMap'	13	D6
EcWantsPipe4'	14	D7
DMadr.09	10	S4
DMadr.10	9	S2
DMadr.11	7	S1

15 MD4

1740

MidasBank3'b	2	E'
1750-1777 unused		

63 ST + A

VictimInST	6	D0
STIdle'	5	D1 i8
StartST	4	D2
STWait-Mem'	3	D3
STState.0	11	D4
STState.1	12	D5
STState.2	13	D6
STState.3	14	D7
DMadr.09	10	S4
DMadr.10	9	S2
DMadr.11	7	S1

15 MD6

1460

MidasBank0'b	2	E'
STfree'	6	D0
VictimInA	5	D1 h21
MapRfshDly	4	D2
RefUsesDInEc1	3	D3
AWordRefToD	11	D4
MapWantsPipe	12	D5
MapFree	13	D6
UseAsrn	14	D7
DMadr.09	10	S4
DMadr.10	9	S2
DMadr.11	7	S1

15 MD7

1470

67 MEMST

MapWait-MemD	6	D0
MapWait-MemIO	5	D1 k15
MemIdle'	4	D2
MemFree	3	D3
MemState.0	11	D4
MemState.1	12	D5
MemState.2	13	D6
MemState.3	14	D7
DMadr.09	10	S4
DMadr.10	9	S2
DMadr.11	7	S1

15 MD6

1560

MidasBank1'b	2	E'
FinNext	6	D0
MemRfsh	5	D1 k22
StopFinTaskLoad	4	D2
DdataGood'	3	D3
MakeSout+D	11	D4
MakeTransport0	12	D5
MakeTransport1	13	D6
MakeTransport2	14	D7
DMadr.09	10	S4
DMadr.10	9	S2
DMadr.11	7	S1

15 MD7

1570

73 EC1 + MAKE

StartEc2'	6	D0
Ec1Free'	5	D1 l14
Ec1Idle	4	D2
Ec1Func.0	3	D3
Ec1Func.1	11	D4
Ec1State.0	12	D5
Ec1State.1	13	D6
Ec1State.2	14	D7
DMadr.09	10	S4
DMadr.10	9	S2
DMadr.11	7	S1

15 MD6

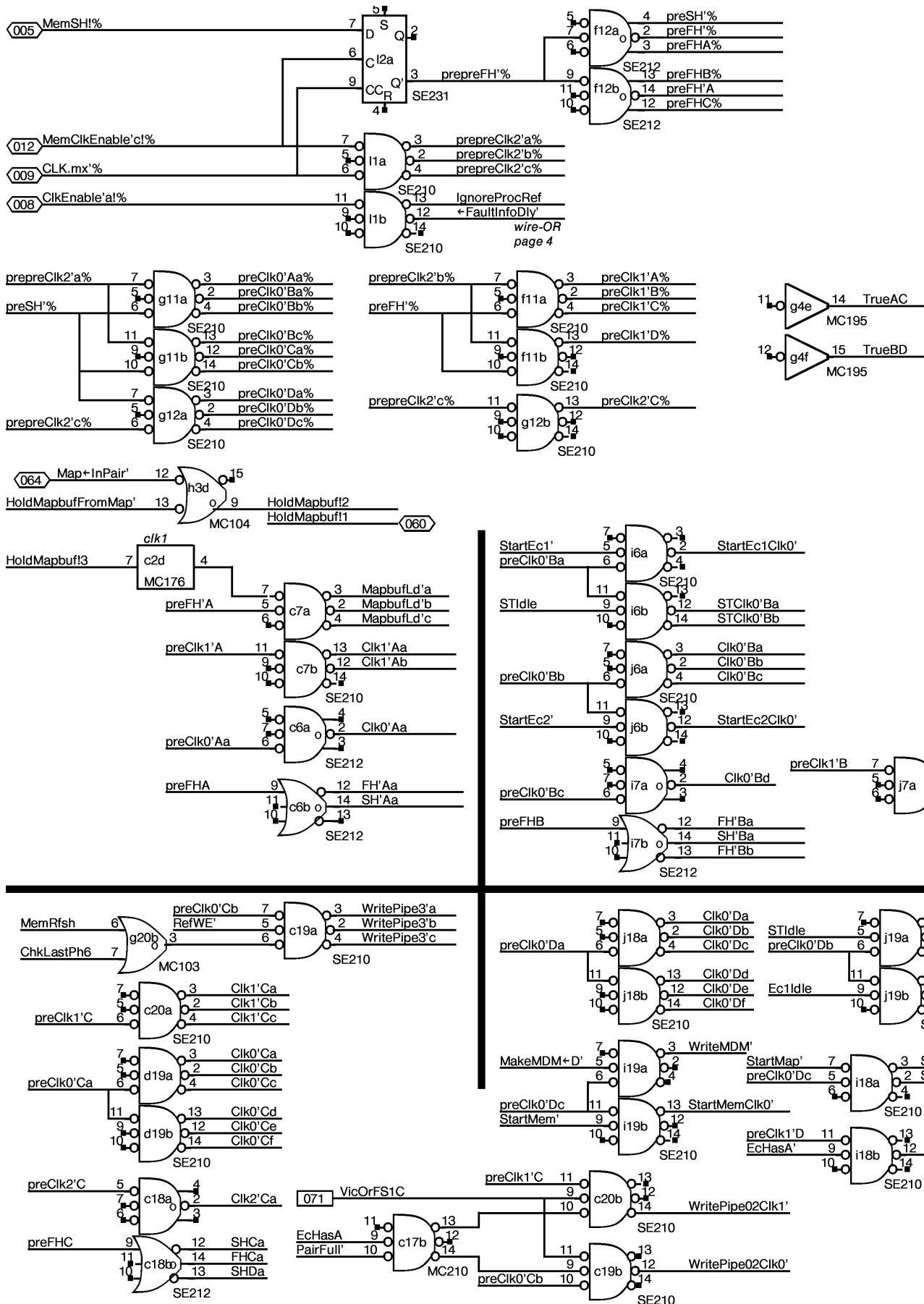
1660

MidasBank2'b	2	E'
EcWantsAa	6	D0
FoutNext	5	D1 k24
MakeFout+D	4	D2
MakeD+CD	3	D3
MakeD+Dbuf	11	D4
MakeF+D	12	D5
MakeMD+D	13	D6
MakeMDM+D'	14	D7
DMadr.09	10	S4
DMadr.10	9	S2
DMadr.11	7	S1

15 MD7

1670

Bank	Addresses
0	1400-1477
1	1500-1577
2	1600-1677
3	1700-1777



A	a 181	b 168	c 153	d 137	e 124	f 109	93 g	80 h	64 i	48 j	33 k	20 l	B
1	BmuxDrivr 0-1	Mapbuf 0-1-8-9	RAS-CAS A,7	Mapbuf 2-3-4-10	BmuxDrivr 12-13	Mapbuf 5-6-7-13	Mapbuf 16,17	BMuxEn 11,11	ErrorRcvr 4,4,10,D	ModSinEn	HOLDcntr	ppclk/IPR	1
41	174 11	175 11	231	175 11	174 11	175 14-15	175 11	109	104	F16 9	F16 8	210 20	
2	BNTGtCT a,4,c,8	BmuxDrivr 8-9	CAS-WE clk1 7,8,20	BmuxDrivr 4-5	BmuxDrivr 6-7	BmuxDrivr 14-15	BMxSel	MU	ShifInEven 15,9,5,5	STOut 5,5,10,10	HOLDcntr	Clock	2
	102	174 11	176 11,11,	174 11	174 11	174 11	103 11	P34INEC	103	102	F16 8	231 20	
3	ModEn	SHIFTS LOADS	BmuxDrivr 10-11	BmuxDrivr 2-3	ModDcode	ModEna	ParityIn	3,3,10,20	clk1 8,4,5,6,5,,	clk0 8,8,10,	A,5,13	Ec2AuIn	3
42	F16 7	103 5,5,5,5	174 11	174 11	171 7	F16 7	170 13	104	176	176 4,4,5	106	176 10	
4	0	MU Mapbuf	MU Mapbuf	11	IN	MCR	10,10,9, 20,20,4 195	dPipe34Ad	dPipe34Ad	MU	MU	MU	4
					125 13	F16 11		174 3	174 3	STOUT	MAPEC2	P34INEC	
5	1	IN	+	12	Mod	A,11,C	RfshCnt	Pipe34Ad	STOut	4,7,11 11,4,4 195	FaultSrn	MidasBank	5
43		105 12			174 8	105	F16 15	141 3	176 5		F16 3	171 16	
6	2	OUT	CLK	13		+	RfshCnt	ProcSrn	CLK	CLK	EcSrn = 0 Fsrn = 0 109 3,3	10,10,10	6
		124 12	212 20				F16 15	141 3	210 20	210 20		105	
7	3	+	CLK	14	OUT	+	RfshCnt	MU	CLK	CLK 20,B	ReportFlt	Ec1AuIn	7
44		210 20		124 13			F16 15	INMAP	212 20	210	121 3	176 9	
8	4	IN		15		+	MU	MemPE	MU	srn0,Wrflsh 135 3,4	MU	Ec1Func	8
		125 13					PEEC	F00 4	ST + A		RFS + SRN	117 9	
9	5	OUT		PAR	IN	+	MU	Ec1Srn	Ec2Srn	LdFltCnt NeedRfsh	FaultCnt	+	9
45		104 10			125 12		APESRN	141 3	141 3	117 3,4	F16 3		
10	6			DTYb	OUT	+	A,8,8,5	PEsrn	MemAuOut	MU	MU	Ec2Func	10
					124 12		104	158 3	F16 7	STOUT	FLT + MEM	F16 10	
11	7	IN	PLAT BYPASS	DTYa	PARITY	CLK	CLK	STProm	7,7,7	MemAuIn	7,7,9 10,10,10 195	Ec2Prom	11
46		125 12			170 13	210 20	210 20	139 4	105	176 7		139 10	
12	8	OUT	PLAT SWITCH	REF	PARITY	CLK	CLK	STState	STProm	MemAuOut	MU	Ec1Prom	12
		124 12			170 13	212 20	210 20	F16 5	139 5	F16 7	FLT + MEM	139 9	
13	9	PLAT BYPASS	WE	WP	ADDR	Pipe34	CheckWP 6,6,6,7 104	MapAu 7,B,6,6 103	MU	MemProm	MemAuIn	MU	13
47			125 14		125 15	145 13			MAPCTRL	139 7	176 7	PEEC	
14	10	PLAT SWITCH	ADDR	RESISTOR	ADDR	Pipe34	MapTrbl	MapAuIn	MapProm	MemProm	clk0 7,7,.... 176	MU	14
			125 15	14	125 15	145 12	121 13	176 6	139 6	139 7		EC1 + MAKE	
15	+	RasCas	ADDR	RESISTOR	ADDR	Pipe34	MapProm	MapState	MapAuIn	STIdle MapWait	MU	+	15
48		125 14	125 15	14	15,15,14 125 14	145 12	139 14	F16 6	176 6	117 7,5	MEMST		
16	+	+	MU	RESISTOR	Pipe34	Pipe34	+Map	MapAdMux	MapAdMux	MemState	13,10,5,9	Ec2State	16
			MAPCTRL	14	145 13	145 12	105 14	158 15	158 15	F16 7	103	F16 10	
17	Pair 2,... F16	+	ASRNInc 3,20	+	OUT	Pipe34	+	MapAdMux	MapWait	VicSTPerr 10,9,C,4 104	Ec1State	6,10,10	17
49			210		124 13	145 13		158 15	121 6		F16 9	104 10	
18	CurTask	MU	CLK	clk1 14,12,, 176 ...	MapRCW	MapRCW	MemCad	MemSrn	CLK	CLK	VicInSt MapWaitD	1,1,9,6	18
	141 2	MCDTSK	212 20		F16 14	F16 14	173 12	141 3	210 20	210 20	109 6,9	104	
19	MU	TrueNext	CLK	CLK	clk0 14,1,1,4 176 f,g	ASRN	MemCad	StartST	CLK	CLK		clk0 5*10,9 176	19
50	MCDTSK	158 2	210 20	210 20		F16 3	173 12	121 5	210 20	210 20	104 10		
20	EMUorFT a,1,1,1 113	MU	CLK	Tag211	ProcTag	MU	12,20,C, 103 13	A,4,4	MapAu	Ec1AuOut	Ec1AuOut	10,9,9 9,6,1 195	20
		TAG + AT	210 20	211 1,B	F145	TAG + AT			104 6	176 9	176 9		
21	Atask	Dt = Ct	PipeTask Pipe2	PipeTaska	PipeSubT Pipe2	MDMTtag	dMDMAd	MU	MU	Ec2AuOut	Ec2AuOut	Midas	21
51	F16 2	113 1	F145 2	197 2	F145 2	F145 1	159 1	ST + A	RFS + SRN	F16 10	F16 10	176 18	
22	dMDMtagAd	At = Ct	FinTask	MDMtagAd	DTask	MakeMD + D	dMDMAd	MapSRN	STOut	MapAUOut	MU	Midas	22
	158 1	113 1	F16 2	141 1	F16 2	117 1	100 1	141 3	176 5	F16 6	MEMST	164 18	
23	ASubTask 2,2,1,1 F16	CurSubT 2,2,d,e 141	RptCur 4,4,4,1 1662	MemAd 0	CAS-WE 1,7,8,4,5,4 176 clk1	1,6,1,D	clk0 1,3,11,e 176 5,5	dPipe02Ad	Hit 1,1,1	Transport 5,5	5,B,1,5	Midas	23
52			159 12		103		158 3	105	212 5,5	102		176 18	
24	bNext 1,1,1,1,... 197	FinSubT 2,2,d,e F16	HOLD	MemAd 1-4	RAS-CAS A,7	CacheRef 1,1,4	Page1 4,1,2 105	Pipe02Ad	MemAd 5-8	FoutTask	MU	Midas 18,18, EC1 + MAKE	24
			1660 4	159 12	231	1674		141 3	159 12	176 2		102 C,18	
C	a 11	b 26	c 39	d 55	e 70	f 86	99 g	114 h	129 i	143 j	159 k	174 l	D
E													E

XEROX PARC	Project Dorado	Reference MemX Board Layout	File MemX21.sil	Designer K. Pier	Rev Cg	Date 10/29/79	Page 21
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Location h11		11 cycles		12 cycles	
ST ADDR		IO Store		VictimStore	
Q STATE		0-13	14-17	20-33	34-37
		012345678910	11	01234567891011	
0	prepreLoadEn'	00000001110	0 1	00000000111 1	1
1	preEcLoadEn'	10101010111	1 1	11010101011 1	1
2	preShiftEn'	00000001111	1 1	10000000111 1	1
3	preEnEcGen'	11111111110	0 1	01111111111 1	1
4	preMapWait-ST'	00001111110	0 1	00000111111 0	1
5	preSTfree'	1111111101	1 1	1111111110 1	1
6	prepreSTPerrNow'	00000000111	1 1	10000000011 1	1
7	preStopFinTaskLd	11111100001	1 0	11111111000 0	0
Location i12					
0	preFinNext	11111110001	1 0	00000000000 0	0
5	preMakeSout+D	00000000000	0 0	11111111000 0	0
6	PreMakeTrnspt0	00000000000	0 0	11111111000 1	0

Location l12		Write	CacheLoad	Clean or Miss	Dirty Hit
EC1 ADDR		0 - 7	10 - 17	IOFetch	IOFetch
Q STATE		01234567	01234567	01234567	01234567
0	preEcEn'	01111111	01111111	01111111	01111111
1	preMakeTrspt2	00000000	00111100	00000000	00111100
2	preMakeFout+D	00000000	00000000	00000000	00011100
3	preFoutNext	00000000	00000000	00011100	00001100
4	preMakeD+CD	00000000	00111100	00000000	00000000
5	preEcWantsA	01000000	01000000	01000000	01000000
6	preEc1Free'	11111101	11111101	11111101	11111101
7	preStartEc2'	11111011	11111011	11111011	11111011

Location l11		Write	CacheLoad	CLEAN or Miss	Dirty Hit
EC2 ADDR		0 - 7	10 - 17	IOFetch	IOFetch
Q STATE		01234567	01234567	01234567	01234567
0	prepreMakeTnspt1	00000000	11100001	00000000	11100001
1	preMakeFout+D	00000000	00000000	00000000	11110001
2	preFoutNext	00000000	00000000	11110001	11111001
3	preMakeD+CD	00000000	11100001	00000000	00000000
4	preEc2Free	00000010	00000010	00000010	00000010
5	preEcWantsPipe4	10000000	10000000	10000000	10000000
6	preMapWait-Ec2	00000000	11110001	00000000	11110001
7	CacheLoad'	11111111	00000010	11111111	11111111

Location i14		Refresh	Read	Write	Map write
MAP ADDR		0 - 7	10 - 17	20 - 27	30 - 37
Q STATE		01234567	01234567	01234567	01234567
0	prepreRfshInMem	00010000	00000000	00000000	00000000
1	UNUSED				
2	preStartMem'	11101111	11101111	11101111	11111111
3	preStartMem	00010000	00010000	00010000	00000000
4	preMapFree'	11111101	11111101	11111101	11111101
5	preMapWantsPipe	00000000	00110000	00110000	00110000

IDENTICAL

Location g15		Refresh	Read	Write	Map Write
MAP ADDR		0 - 7	10 - 17	20 - 27	30 - 37
Q STATE		01234567	01234567	01234567	01234567
0	preMemRfsh	00011110	00000000	00000000	00000000
1	preRAS1'	00000110	00000110	00000110	00000110
2	preRAS2'	00001000	00001000	00001000	00001000
3	prepreCAS1'	11111110	00000010	00000010	00000010
4	prepreCAS2'	11111100	00000000	00000000	00000000
5	preMapWE'	11111111	11111111	11111111	11100111
6	preRefWE'	11100111	11100111	11100111	11100111
7	preDirtyWE'	11111111	11111111	11100111	11100111

Location j13		READ	WRITE	REFRESH
MEM 16K chips		0 - 7	0 - 7	10-17
Q		01234567	01234567	01234567
0	prepreMemWE'	10011111	10011111	1 1111111 1
1	preSTWait-Mem'	10111111	10111111	1 1111111 1
2	prepreMemCAS	11111100	11111100	0 0000000 0
3	prepreShiftLoadEven	00010000	00010000	0 0000000 0
4	prepreMCS	11111000	11111000	0 0000000 0
5	preMemFree	00000010	00000010	0 0000001 0
6	preMemState6'	11111011	11111011	1 1111101 1
7	preMemRAS	11110001	11110001	1 1111001 1

IDENTICAL

Location j14		READ					WRITE					REFRESH					
MEM 4K chips		0 - 14					15-17					20 - 34					35-37
Q	ADDR STATE	0123456789	10	11	12	13-15	0123456789	10	11	12	13-15	0123456789	10	11	12	13-15	
0	prepreMemWE'	11000000111	1	1	1	1	11000000111	1	1	1	1	1111111111	1	1	1	1	
1	preSTWait-Mem'	1111101111	1	1	1	1	1111101111	1	1	1	1	1111111111	1	1	1	1	
2	prepreMemCAS	0001111100	0	0	0	0	0001111100	0	0	0	0	0001111100	0	0	0	0	
3	prepreShiftLoadEven	0000000100	0	0	0	0	0000000100	0	0	0	0	0000000000	0	0	0	0	
4	prepreMCS	0111111000	0	0	0	0	0111111000	0	0	0	0	0000000000	0	0	0	0	
5	preMemFree	0000000000	0	1	0	1	0000000000	0	1	0	1	0000000000	0	1	0	1	
6	prepreState7'	1111111111	1	1	1	1	1111111111	1	1	1	1	1111111111	1	1	1	1	
7	preMemRAS	1111111100	0	0	1	0	1111111100	0	0	1	0	1111111100	0	0	1	0	

IDENTICAL

XEROX	Project	Reference	File	Designer	Rev	Date	Page
PARC	Dorado	Automata program tables	Memx22.sil	K. Pier	Cg	9/08/79	22

**Dorado Memory Extension Board
Stuffing and Configuration Instructions**

1. If 256 chips are NOT installed in the Memory Storage Arrays (MSA), then
 - a. Break g10.6 (MC104) before stuffing. Label this chip as MemX-g10.
2. Break h20.10 before stuffing. Label this chip as MemX-h20.
 All SG10139 chips are PROMS which must be blown and labeled before stuffing.
 Stuff all PLAT1816 with Beckman type 898-3-R27 resistor packs or equivalent.
3. Do all the actions in one of the following three columns:

Action	16K Map Chips	64K Map Chips	256K Map Chips
Blue Wire	k4.6 to k4.16	k4.5 to k4.16	k4.4 to k4.16
Stuff into MosRam sockets	MK4116-2 or equivalent	TMS4156 or equivalent	256K Ram or equivalent
Jumper wires in socket b14	3 to 14 5 to 12 7 to 10	6 to 11 4 to 13	6 to 11 4 to 13 2 to 15
Jumper wires in socket c12	3 to 14 5 to 12 7 to 10	6 to 11 4 to 13	6 to 11 4 to 13 2 to 15
capacitors in SIP sockets b42,b43,b44,b45,b46,b47 d42,d43,d44,d45,d46,d47	0.1 mmf pins 5 to 8 each socket AND 0.1 mmf pins 1 to 4 each socket	0.1 mmf pins 5 to 8 each socket	0.1 mmf pins 5 to 8 each socket
1 KOhm resistor in socket b14 and 1 KOhm resistor in socket c12	NO	1 to 15 IF Ram has self refresh feature- Motorola or equivalent	NO
22 uf Cap in PLAT c11 & b13	+ 2 to 15 + 4 to 14 + 11 to 6	+ 3 to 14 + 5 to 12	+ 3 to 14 + 5 to 12

